

A-level

# ENVIRONMENTAL SCIENCE

## Questions by topic: Habitat case studies: woodland, rainforest, reefs, islands, mangroves, Antarctica - Set 1

### Mark scheme

AQA spec 3.1.2.3.4; The living environment; textbook pp. 50-71

### Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

**Textbook references:** every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Marks |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 01.1     | <ul style="list-style-type: none"> <li>• No major temperature extremes; regular water supply with no pronounced dry season; deep fertile soils built by leaf litter and root binding; strong seasonality</li> <li>• Deciduous canopy present spring to autumn shades the floor during the best growing temperatures</li> <li>• Bluebells, snowdrops, lesser celandine grow and flower early in spring before the trees leaf</li> <li>• Winter food shortage causes animals to hibernate (dormice), store food or migrate</li> </ul> | 4     |
| 01.2     | <ul style="list-style-type: none"> <li>• Long time for species to colonise; some woodland plants colonise very slowly so their presence (ancient woodland indicators) shows the wood is old</li> <li>• Natural woodland has trees of all ages with clearings where old trees died, giving varied light and dead wood habitats</li> <li>• A new plantation has a single-age dense canopy with low light and few niches until trees start to die (over 200 years)</li> </ul>                                                          | 3     |
| 01.3     | <p><i>Any 3 from:</i></p> <ul style="list-style-type: none"> <li>• Clearance for farmland, plantations, urban development, roads, mineral extraction</li> <li>• Fragmentation isolating remaining woods so populations cannot move between them and are vulnerable</li> <li>• Management change: end of coppicing so woods become uniform and dark; replacement by conifer monocultures</li> <li>• Overgrazing by deer preventing regeneration; introduced pathogens (ash dieback)</li> </ul>                                       | 3     |
|          | <p><b>Source:</b> R. Genn, Environmental Science A level AQA (Insight &amp; Perspective, 2018), pp. 50-52; AQA spec 3.1.2.3.4.</p>                                                                                                                                                                                                                                                                                                                                                                                                  |       |

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Marks |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 02.1     | <ul style="list-style-type: none"> <li>• A: coppice rotation creates a patchwork of different ages / light levels - clearings with wildflowers for butterflies, dense regrowth for nesting birds; oak supports many invertebrates</li> <li>• B: same tree species but the closed canopy shades the floor so few flowering plants, fewer invertebrates and butterflies; fewer niches</li> <li>• C: non-indigenous single-species, single-age conifers support few native invertebrates; dense shade; no clearings or dead wood; needles acidify soil</li> <li>• Habitat diversity / structural variety increases the range of niches and hence species</li> </ul>                                                                                                                        | 5     |
| 02.2     | <p><i>Any 3 from:</i></p> <ul style="list-style-type: none"> <li>• Designated protected areas: SSSIs, NNRs (Sherwood Forest), SACs (Epping Forest), National Parks</li> <li>• Legal protection of ancient woodland: planning permission refused for developments destroying irreplaceable habitats (2012 NPPF)</li> <li>• Conservation management: coppicing, creating clearings, planting mixed native species, leaving dead wood, controlling deer</li> <li>• Planting new native woodlands linking fragments (corridors); Woodland Trust / National Trust ownership</li> </ul>                                                                                                                                                                                                       | 3     |
| 02.3     | <ul style="list-style-type: none"> <li>• No species is completely dominant and food webs have many alternative links</li> <li>• So a change in the population of one species has a relatively small impact on the whole community</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2     |
|          | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 50-52; AQA spec 3.1.2.3.4.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |       |
| 03.1     | <ul style="list-style-type: none"> <li>• Stable tropical climate over long periods gave time for species to evolve and adapt to local abiotic and biotic factors; many rare species with small ranges</li> <li>• High light, warm temperatures and high rainfall all year allow continuous photosynthesis and growth, supporting rich food webs with no seasonal food shortage (no hibernation / migration)</li> <li>• Stratification (emergent, canopy, understorey, shrub, ground layers) and epiphytes create many niches; intense competition for light drives specialisation</li> <li>• Biotic pressures (predation, inter-species relationships) drive evolution of specialised pollination / seed dispersal relationships; plants flower and fruit at different times</li> </ul> | 4     |
| 03.2     | <ul style="list-style-type: none"> <li>• Adapted to stable conditions so they have narrow ranges of tolerance; a small change can move conditions outside the range</li> <li>• Many species are rare with small ranges / small populations</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2     |
| 03.3     | <ul style="list-style-type: none"> <li>• No cold winter kills herbivores so plants defend themselves with toxic chemicals (alkaloids) that are medicinally active in controlled doses (quinine, diosgenin, curare)</li> <li>• Animals without teeth / claws (frogs, reptiles) rely on chemical defences or venoms (fer-de-lance venom - captopril)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                           | 2     |
| 03.4     | <ul style="list-style-type: none"> <li>• Over half the Amazon's rainfall comes from transpiration within the forest; clearance reduces downwind rainfall</li> <li>• Rapid decay recycles nutrients so soils are shallow / poor; canopy and roots prevent erosion of the thin soil - clearance leads to rapid erosion and infertility</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2     |

| Question    | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Marks |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 52-55; AQA spec 3.1.2.3.4.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |       |
| <b>04.1</b> | <ul style="list-style-type: none"> <li>• Small clearings were farmed for a few years then abandoned as nutrients declined; the forest recolonised and soil fertility recovered before the area was cleared again</li> <li>• Population growth shortens the gap between clearances so the forest and soil do not recover</li> <li>• Clearings enlarge and degrade; less regrowth</li> </ul>                                                                                                                                                                                                                                                             | 3     |
| <b>04.2</b> | <ul style="list-style-type: none"> <li>• Rainforest soils are thin and nutrient-poor (nutrients held in the biomass); once cleared, fertility falls in a few years and yields drop, so more land is cleared</li> <li>• Abandoned land may not be recolonised if no forest survives nearby; erosion of exposed soil</li> </ul>                                                                                                                                                                                                                                                                                                                          | 2     |
| <b>04.3</b> | <ul style="list-style-type: none"> <li>• Mining (aluminium, iron, gold, copper) destroys habitat directly and pollutes rivers with turbid drainage, cyanide and mercury from processing</li> <li>• Reservoirs (eg Tucurui, 2850 km<sup>2</sup>) flood large areas, block migratory fish and river dolphins; decomposing vegetation produces acidic, deoxygenated water; fluctuating downstream flows</li> <li>• Droughts increase frequency, area and duration of fires (long fires boil sap and kill trees); cloud forests lose humidity as clouds form higher so frogs and epiphytes decline; some models predict replacement by savannah</li> </ul> | 3     |
| <b>04.4</b> | <ul style="list-style-type: none"> <li>• A creditor country or conservation charity cancels / buys part of a debtor country's foreign debt</li> <li>• In return the country guarantees to protect an area of rainforest (eg USA - Guatemala, Brazil; WWF - Costa Rica, Philippines)</li> </ul>                                                                                                                                                                                                                                                                                                                                                         | 2     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 55-57; AQA spec 3.1.2.3.4.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |       |

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Marks |
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| 05.1     | <p><b>Levels of response</b></p> <p><b>Level 3 (7-9 marks):</b> Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached.</p> <p><b>Level 2 (4-6 marks):</b> Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported.</p> <p><b>Level 1 (1-3 marks):</b> Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation.</p> <p><b>0 marks:</b> No relevant content.</p> <p><b>Indicative content</b> (credit other relevant, accurate points)</p> <ul style="list-style-type: none"> <li>• Protected areas: Korup NP Cameroon (1260 km<sup>2</sup>, 600 tree species, endemic <i>Ancistrocladus korupensis</i> with anti-HIV / antimalarial alkaloids); Tumucumaque NP Brazil (39 000 km<sup>2</sup>, adjoining French Guiana reserve); Alto Maues; Rio Bravo Belize funded by charities, companies and ecotourism</li> <li>• Debt for nature swaps; sustainable exploitation: selective logging of high-value timber (mahogany) funding conservation; ITTO aims (largely unmet); CITES listing of timber species (Honduras mahogany Appendix II)</li> <li>• Ecotourism income (but visitor damage in Costa Rica, Malaysia); community projects and employment; carbon sequestration payments</li> <li>• Effectiveness: protected areas work where enforced but face illegal logging, mining, poaching, roads, conflict; drivers (population growth, demand for beef, soya, palm oil, timber, minerals) originate outside the forest; climate change cannot be controlled locally</li> <li>• Judgement: combination of large protected areas, economic alternatives for local people and reduced international demand needed; overall loss continues</li> </ul> | 9     |
| 05.2     | <ul style="list-style-type: none"> <li>• Carbon sequestration reducing climate change; genetic resources / medicines; rainfall regulation; global biodiversity</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 1     |
|          | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 56-57; AQA spec 3.1.2.3.4.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |       |
| 06.1     | <ul style="list-style-type: none"> <li>• Zooxanthellae are photosynthetic algae living inside polyp cells (up to 30 000 per mm<sup>2</sup>)</li> <li>• They use the polyp's CO<sub>2</sub> and excretory wastes (nitrogen, phosphorus) for photosynthesis and growth, so nutrients are recycled efficiently</li> <li>• They supply the polyp with glucose and oxygen - over 95% of its food energy; the polyp provides a protected habitat and nutrients</li> <li>• Polyps also trap plankton with cilia and stinging nematocysts; surplus CO<sub>2</sub> is used to build the calcium carbonate skeleton</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 4     |
| 06.2     | <ul style="list-style-type: none"> <li>• High light (shallow, clear water) for the algae; warm stable temperatures (about 18-30 °C); low turbidity; constant salinity; well-oxygenated water</li> <li>• Found only where all these coincide (about 1% of ocean area)</li> <li>• Small changes (warming, sediment, freshwater runoff) move conditions outside the narrow range of tolerance</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 3     |

| Question    | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Marks |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>06.3</b> | <p><i>Any 3 from:</i></p> <ul style="list-style-type: none"> <li>• Fisheries: support over a quarter of marine species including fish, lobsters, conch</li> <li>• Erosion protection: complex structure dissipates wave energy protecting low coral islands; coral sand builds shores</li> <li>• Medicinal discoveries: AZT and Ara-C from sponges, dolastatin from sea hares, bryostatin from Bugula, trabectedin from sea squirts, secosteroids, coral for bone grafts</li> <li>• Climate control: carbonate skeletons sequester CO<sub>2</sub></li> <li>• Tourism (diving, snorkelling)</li> </ul>       | 3     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 57-60; AQA spec 3.1.2.3.4.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |       |
| <b>07.1</b> | <ul style="list-style-type: none"> <li>• Overall decline from 62% to 22%</li> <li>• Sharp falls in 1998 (to 35%) and 2016 (to 18%) coinciding with bleaching events, with partial recovery after each</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                            | 2     |
| <b>07.2</b> | <ul style="list-style-type: none"> <li>• The polyps expel their symbiotic zooxanthellae so the coral loses colour; bleached corals are alive but grow very slowly and may die if algae do not return</li> <li>• Raised water temperature (climate change / El Niño)</li> <li>• Pollution: sewage, pesticides, cyanide, sunscreen; low oxygen from zooplankton blooms after overfishing</li> <li>• High light / solar activity; ocean acidification hinders skeleton formation</li> </ul>                                                                                                                    | 4     |
| <b>07.3</b> | <ul style="list-style-type: none"> <li>• Coastal construction and dredging increase sedimentation / turbidity covering and killing polyps</li> <li>• Sewage and fertiliser nutrients cause algal growth that shades corals and prevents egg release</li> <li>• Physical damage from anchors, divers, souvenir collection; loss of mangroves / seagrass nurseries that trap sediment</li> </ul>                                                                                                                                                                                                              | 3     |
| <b>07.4</b> | <ul style="list-style-type: none"> <li>• Protected breeding populations grow larger and produce surplus young / adults that spill over into fished areas</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                         | 1     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 60-63; AQA spec 3.1.2.3.4.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |       |
| <b>08.1</b> | <ul style="list-style-type: none"> <li>• Giant tritons are predators of the crown-of-thorns starfish which eats coral polyps</li> <li>• Removing tritons allowed starfish populations to explode, destroying large areas of reef (food web / trophic cascade)</li> </ul>                                                                                                                                                                                                                                                                                                                                    | 2     |
| <b>08.2</b> | <ul style="list-style-type: none"> <li>• Fixed mooring buoys so boats do not anchor on coral</li> <li>• Education of divers / snorkellers not to touch coral or kick up sediment; bans on spear fishing and souvenir collection</li> <li>• Fishing controls: no-take zones (Chagos, Great Barrier Reef, Lundy, Lamlash Bay), seasons, minimum and maximum catch sizes, bans on dynamite / small-mesh nets / trawling</li> <li>• Protection of turtle nesting beaches; control of coastal development; protection of mangroves and seagrass; artificial reefs (reef balls, sunk ships / oil rigs)</li> </ul> | 4     |
| <b>08.3</b> | <ul style="list-style-type: none"> <li>• No sunlight so no symbiotic algae; polyps rely only on captured food so have far less energy and grow very slowly</li> <li>• Damage from trawls takes hundreds of years to repair; the slow-growing, low-fecundity fish (orange roughy) they support are easily over-exploited</li> </ul>                                                                                                                                                                                                                                                                          | 2     |

| Question | Marking guidance                                                                                                                                                                                                                                                  | Marks |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 08.4     | <ul style="list-style-type: none"> <li>• Deep-water trawling (also by-catch); oil and gas exploration in deeper water; ocean acidification</li> <li>• Protected areas banning trawling, eg Darwin Mounds SAC off NW Scotland, Oculina Bank off Florida</li> </ul> | 2     |
|          | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 61-64; AQA spec 3.1.2.3.4.                                                                                                                                           |       |